BiDex-100: Supplementary Material

This supplement documents (A) the goal-predicate catalog used by the evaluator, (B) the task-specification and scoring semantics, (C) the out-of-domain transformation rules, (D) the dataset structure and quality-control procedure, (E) per-model training configurations for the seven evaluated VLAs, (F) the evaluation protocol, and (G) the complete task catalog with per-task instructions, image, and scoring criteria, and (H) discussion and limitations.

I. Goal-Predicate Catalog

Table I lists the 42 canonical predicate types used by the 106 released base task specifications. The evaluator additionally accepts backward-compatible aliases for several of these types. A predicate $\phi(\mathbf{s}_t; \text{args}, \epsilon)$ maps the simulator state to $\{0, 1\}$; *args* name objects, regions, articulated joints, or event counts, and ϵ is a per-task tolerance. Selected negated guard predicates may adopt *latch-violation* semantics (Appendix II), under which a forbidden state permanently invalidates the corresponding subtask.

TABLE I: Goal-predicate catalog used by the 106 released base task specifications (42 canonical types).

Predicate	**Semantics (tolerance ϵ unless noted)**
Spatial placement	
`on`	object center within a target's horizontal radius and above it in height
`stacked_on`	object one layer above another within horizontal and vertical windows
`on_table`	object directly supported from below by the scene table
`inside`	object center within a circular horizontal radius and a vertical window
`inside_contact`	upward support contact from a container plus a minimum height
`inside_box`	object center inside a container's world-axis box footprint and height window
`inside_site`	object inside a live named site volume with contact gating
`inside_site_loose`	live named-site containment without the contact requirement
`inside_site_any`	contact-gated containment in any listed named site
`inside_site_any_loose`	containment in any listed named site without contact gating
`inside_region`	object center inside a live named three-dimensional region
`not_inside_region`	exact negation of `inside_region`
`keypoint_in_site`	a selected endpoint of an elongated object inside a named site
`supported_in_region`	live-region containment together with upward support contact
`axis_in_region`	any sampled point on an object's long axis inside a live three-dimensional region
`supported_in_region_band`	region-band containment together with upward support contact
`supported_geom_in_region_band`	named object-geometry center in a region band with upward support
`supported_in_local_box`	explicit container-local box containment with upward support
`inside_local_box`	explicit container-local box containment without contact gating
`inside_any`	`inside` holds for at least one listed container
`not_inside`	exact negation of `inside`
`hung_on`	raised upward-support contact on a holder, excluding its base
`lifted`	object height above an absolute or initialization-relative threshold
`not_lifted`	exact negation of `lifted`
`upright`	object's local z-axis sufficiently aligned with world vertical
`long_axis_vertical`	an elongated object's longest geometry axis sufficiently aligned with vertical
`on_edge`	object's local z-axis approximately horizontal
Articulation	
`open`	joint past a threshold toward the range endpoint farther from zero
`open_any`	`open` holds for any listed articulated joint
`closed`	exact negation of `open`
State-change and event predicates	
`sunk`	nail displaced downward from its initial height by a required depth
`not_sunk`	exact negation of `sunk`
`watered`	actuated spray nozzle currently within the plant's spatial window
`not_watered`	exact negation of `watered`
`pressed_exact`	environment-recorded button count exactly n
`pressed_after`	exact press count whose recorded events all follow a prerequisite count
`in_bottle`	object inside a live bottle-relative radial and height window
`not_in_bottle`	exact negation of `in_bottle`
Counting and memory predicates	
`swaps_n`	environment-recorded color swaps at least n
`color_match_place`	object placed in a container whose current color matches its remembered color
`placed_n`	at least n first-entry placement events recorded for a target
`order_match`	all orderable objects removed in their recorded placement order

II. Task Specification and Scoring Semantics

a) Task structure.: Every task specifies a robot embodiment and scene, a set of placed objects with their spawn regions, a natural-language instruction, a goal composed of typed predicates, and a list of scored subtasks. Each spawn region defines a center, an xy-extent, and a yaw range; at reset, every assigned object is placed at a pose sampled uniformly from its region, so initial conditions differ across episodes even within one seed family. A surface-contact flag places objects resting on the support surface rather than dropping them.

b) Subtask latching and scoring.: At each control step all predicates are evaluated, and subtask k latches the first time all of its predicates hold simultaneously. Let $\ell_k(T)$ denote this latched state, m_k indicate that the subtask has a must-hold predicate, $h_k(T)$ denote its terminal truth, and $v_k(T)$ indicate a latched violation. Its effective completion is $e_k(T) = \ell_k(T)(1 - m_k + m_k h_k(T))(1 - v_k(T))$. Strict success is $\prod_k e_k(T)$ and the progress score is $\sum_k w_k e_k(T)$, where $w_k \geq 0$ and $\sum_k w_k = 100$. Zero-weight safety or terminal constraints still participate in strict success. For ordered tasks, the evaluator permits a subtask to latch only after the required earlier subtasks or groups have latched, enforcing the intended staging.

c) Language schedules.: Instruction-Following *switch* tasks declare an ordered schedule of instruction segments. Each nonfinal segment specifies an `until_step` control-step boundary, while the last segment supplies the instruction for the remainder of the episode. The policy observation receives the active segment text. Scoring continues to use the task's declared predicate set, with ordered subtasks enforcing the intended sequence.

d) Episode horizon.: A task-level `max_steps` value takes precedence when present. Otherwise, the evaluation horizon is $1.2\times$ the 90th-percentile demonstration length in control steps, with a 30 s fallback when no demonstration lengths are available. Timeouts retain the latched partial score.

III. Out-of-Domain Transformation Rules

Each in-domain task is paired with a single out-of-domain (OOD) variant that perturbs the scene but keeps the goal unchanged, so the OOD score isolates robustness rather than task mastery. Transformations are deterministic and reproducible across runs and machines through stable, component-scoped keys. Six transformation families are used:

1) **Scene randomization** (all tasks): floor, wall, and tabletop textures resampled from a held-out pool; light position and diffuse intensity jittered; shadow casting toggled.
2) **Camera variation** (all tasks): head and wrist camera extrinsics selected from a validated finite pose bank for each evaluation key.
3) **Layout variation** (all tasks): task-specific placement and geometry changes are recorded in the committed OOD YAML.
4) **Distractor injection** (45 compatible tasks): three to nine objects are placed in collision-safe tabletop regions; grasping a distractor terminates the episode at its current score, so OOD scores penalize both distraction and mis-grasping.
5) **Object rescaling** (ten articulated Limited-Demonstrations tasks): each OOD YAML selects a smaller or larger regime anchored by a factor in $\{0.85, 0.9, 1.1, 1.15\}$. A deterministic episode factor is sampled within ± 0.18 of the anchor and clipped to $[0.65, 0.96]$ for smaller regimes or $[1.04, 1.35]$ for larger regimes.
6) **Left-right mirroring** (nine tasks): task geometry is mirrored about the workspace midline to exchange arm-side demands.

At evaluation time the OOD variant is loaded in place of the base task. The committed YAML fixes the layout and applicable transformations, while appearance, camera, articulated scale, and distractor placement are derived reproducibly from the task, seed, and episode. The goal, episode count, horizon, and scoring rules are shared across domains. In-domain evaluation samples a recorded working robot pose and restores only the robot joints, while retaining fresh base-YAML object placement. OOD evaluation uses the default robot reset pose with the committed OOD placement. Each domain is deterministic and identical across models.

IV. Dataset Structure and Quality Control

a) Layout.: The dataset is released in LeRobot v2.1 and v3.0 layouts, one directory per task, storing per-episode state/action trajectories and six synchronized camera videos. Table III summarizes the observation and action modalities available to the training and evaluation interfaces.

b) Quality control.: Collection proceeds as teleoperated sessions with an in-loop review interface: each episode is recorded, replayed, and accepted or rejected by the operator. Accepted episodes are then re-validated automatically by replaying them from their serialized initial states and re-checking the task's goal predicates; only episodes that re-achieve full subtask completion are exported. The released corpus is therefore success-only by construction. Each task additionally ships a recorded initial state per episode (state dimension 120–222, depending on the object set), which enables exact replay and fully deterministic evaluation resets.

c) Normalization statistics.: Per-task and per-mixture normalization statistics (mean/std and 1st/99th percentiles of every action and observation channel) are distributed with the dataset and used by all baselines (Appendix V).

TABLE III: Observation and action modalities per frame (25 Hz). The six video streams are RGB and depth from the head and the two wrist cameras.

Modality	Content	Dim
Observations		
Arm end-effector pose (right, left)	xyz + quaternion, world or head-camera frame	14
Arm joint position / velocity	7 per arm	14 each
Hand joint position / velocity	22 per hand	44 each
Actions (one 58-dimensional control representation)		
Arm component	joint targets or end-effector pose in world/head-camera frame	14
Hand joint targets	22 per hand	44
Video streams (H.264, 25 fps)		
Head / left-wrist / right-wrist RGB		360×480
Head / left-wrist / right-wrist depth	3-channel	360×480
Per-episode		
Simulator initial state	for exact replay / deterministic reset	120–222

TABLE IV: Training hyperparameters and selected checkpoints of the seven evaluated VLAs (cosine-family schedules, grad-clip 1.0). GBS = global batch size.

Model	Train	Optimizer	Peak LR	Warmup	Min LR	GBS	Dtype	Chunk	Frozen modules
$\pi_{0.5}$	60k	AdamW (optax)	5e-5	100	5e-5	256	bf16	50	none (EMA 0.999)
GR00T-N1.7	400k	AdamW	1e-4	1500	$\sim$0	256	bf16	16	VLM + vision encoder
InternVLA-A1.5	400k	AdamW	5e-5	2000	5e-6	256	bf16	50	none
Galaxea G0.5	400k	AdamW	1e-5	1500	1e-6	256	bf16	32	none
OpenDM-DM05	120k	AdamW	2.5e-5	1000	2.5e-6	256	bf16	50	VLM token embeddings
LingBot-VLA-v2	60k	Muon (+AdamW)	1e-4	0	5e-5	256	fp32	50	none
Xiaomi XR1	400k	FusedAdam	2e-5	500	5e-6	128	bf16	30	VLM input embeddings

V. Per-Model Training Configurations

All seven baselines are fine-tuned from their official pretrained checkpoints on the identical all-category BiDex-100 mixture (58-DoF absolute joint targets; head + left-wrist + right-wrist color streams) using the checkpoint selected for the main evaluation: 60k steps for $\pi_{0.5}$ and LingBot-VLA-v2, 120k for OpenDM-DM05, and 400k for GR00T-N1.7, InternVLA-A1.5, Galaxea G0.5, and Xiaomi XR1. Table IV summarizes the optimizer settings; model-specific notes follow.

a) $\pi_{0.5}$ (openpi).: PaliGemma-2B VLM with a 300M-parameter flow-matching action expert; JAX/pmap training in bf16. Uses *relative* (delta) 58-DoF actions; action horizon 50 with 10 denoising steps at inference (single training noise step, $t \sim \mathrm{Beta}(1.5, 1)$). The pretrained 32-DoF action projections are resized to 58 DoF by re-initializing mismatched layers. Images resized to 224×224.

b) GR00T-N1.7.: Cosmos-Reason2-2B VLM (Qwen3-VL architecture) with a 16-layer Alternate-VL DiT flow-matching expert (hidden 1024). The VLM and vision tower are frozen; only the DiT, state/action projectors, and the VLLN adapter train (DeepSpeed ZeRO-2). Absolute 58-DoF actions; chunk 16, 4 inference denoising steps; color-jitter augmentation; min/max percentile normalization.

c) InternVLA-A1.5.: Qwen3.5-2B with a 1B-parameter action expert; flow-matching MSE weighted $\times10$. The pretrained 32-DoF action projection is trimmed and rebuilt at 58 DoF. Chunk 50, 10 denoising steps.

d) Galaxea G0.5.: Qwen3.5-2B (mrope + FLA linear attention) with a flow-matching action expert (1024-wide, adaLN). Absolute 58-DoF actions; chunk 32, single-step FM Euler at inference. The base 27-DoF action projections are partially copied and zero-padded to 58 DoF; the RVQ action tokenizer is bypassed in flow-matching mode. Learning rate set to $1/8$ of the model's default (1e-5) with a warmup-constant-cosine schedule.

e) OpenDM-DM05.: Gemma3-4B-IT + SigLIP-SO400M (448×448, 256 visual tokens per image) with an independent Gemma3-1B action expert (rectified flow). Absolute actions padded from 32 to 64 slots (58 active); chunk 50, 10 denoising steps. VLM token embeddings frozen; eager attention; Liger kernels; gradient checkpointing.

f) LingBot-VLA-v2.: Qwen3-VL-4B with a 36-layer Mixture-of-Experts flow-matching expert (32 experts, top-4) trained with an L1-FM loss. Muon on 2D+ weights (routed-expert LR $\times\sqrt{32/4}$) with AdamW for 1D/embeddings; full fp32 training. Absolute 58-DoF actions (55-DoF pretrained projection resized); chunk 50, 10 denoising steps; alignment/depth auxiliary heads disabled.

g) Xiaomi XR1.: Qwen3-VL-4B with a 36-layer 60-slot DiT expert (AdaLN, cross-attention to VLM key-values); the 58 active channels occupy the right-aligned slots. FusedAdam with a LambdaLR cosine schedule; chunk 30, 5 denoising steps; training repeats each noise sample $4\times$. VLM input embeddings frozen; vision encoder and language FFNs train with gradient checkpointing; DeepSpeed.

VI. Evaluation Protocol

a) Protocol.: Each (model, task, domain) triple is evaluated over 2 fixed seeds $\times$ 25 episodes, using ID seeds 0 and 1 and OOD seeds 2 and 3. Object placements are sampled from the applicable task regions with a seed that is a deterministic function of the task, seed index, and episode index. The ID protocol additionally restores a deterministically selected working robot pose, while OOD uses the default robot reset pose. This procedure makes every rollout reproducible within its domain. Episodes terminate on full success, on timeout (explicit task `max_steps`, otherwise $1.2\times$ the P90 demonstration length), or, in OOD scenes with distractors, on grasping a distractor object. The vectorized harness co-locates policy inference and simulation with a single shared renderer per worker.

b) Metrics.: Per episode we record strict success, the progress score (as defined in the main paper), the per-subtask latch times, and the time-to-first-failure. Aggregation reports, per capability axis and domain: mean strict success (*SR*) followed by mean progress score (*Score*); the overall number is the equal-weight mean over all 212 task-domain means. Per-task breakdowns and the full rollout videos are retained for audit, and the leaderboard requires no human labeling.

VII. Task Catalog

This section documents every one of the 106 task specifications, grouped by capability axis. Each axis opens with a short description of the capability it stresses, followed by one entry per task. Each entry shows a representative head-camera image of the task scene, the task identifier, the number of scored subtasks K, the number of released demonstrations, the language instruction, and the full scoring criteria, namely the subtask names together with their integer score weights, which sum to 100 per task. Ordered subtask chains are joined by "$\rightarrow$" and unordered subtasks by "$\cdot$". A subtask marked † carries *must-hold* predicates that must still be true at episode end; ‡ marks *latch-violation* predicates that permanently fail the episode if the forbidden state ever occurs (Appendix II). Ten Instruction-Following shelf-transfer switch and combination specifications contain 30 demonstrations each, while the other ten Instruction-Following specifications contain 100 demonstrations each.

A. Generalization (40 specifications, 40 with released demonstrations)

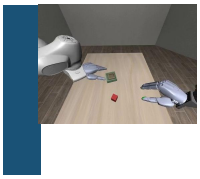
Single- and few-stage skills, including lifting, placement, insertion, threading, and cooking, over a diverse object library spanning food items, tableware, containers, and tools. The ten `kitchen_*` tasks execute on the Astribot S1 humanoid morphology and the remaining 30 use the dual FR3v2 platform; all 40 are included in the reported Generalization results. In-domain resets already randomize object poses, so the axis measures transfer across object geometry and embodiment.



1. `insert_two_cubes_into_slot` — K=2 subtasks · 100 demos

Instruction: Insert two cubes into the slot.

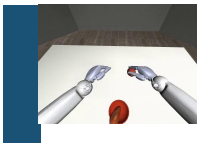
Scoring: Insert cube A into the slot **(50)**† · Insert cube B into the slot **(50)**†.



2. `nest_medium_box_in_large_box` — K=4 subtasks · 100 demos

Instruction: Place the medium box inside the large box, then place the cube inside the medium box.

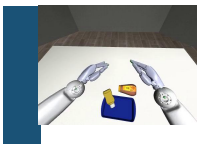
Scoring: Pick up the medium box **(10)** · Nest the medium box inside the large box **(40)**† · Pick up the cube **(10)** · Place the cube inside the medium box **(40)**†.



3. `put_sweet_potato_and_onion_in_bowl` — K=3 subtasks · 100 demos

Instruction: Put the sweet potato and the onion in the bowl.

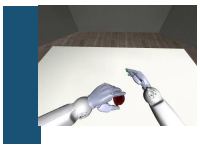
Scoring: Pick up the sweet potato **(25)** · Put the sweet potato in the bowl **(25)**† · Put the onion in the bowl **(50)**†.



4. `put_mustard_and_honey_bottle_on_tray` — K=3 subtasks · 100 demos

Instruction: Put the mustard and the honey bottle on the tray.

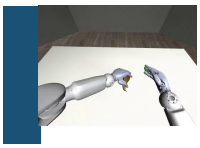
Scoring: Pick up the mustard **(25)** · Place the mustard on the tray **(25)**† · Place the honey bottle on the tray **(50)**†.



5. `pick_up_peach` — K=1 subtask · 100 demos

Instruction: Pick up the peach.

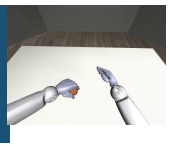
Scoring: Pick up the peach **(100)**.



6. `pick_up_pear` — K=1 subtask · 100 demos

Instruction: Pick up the pear.

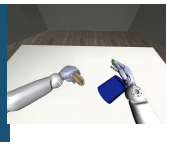
Scoring: Pick up the pear **(100)**.



7. `pick_up_tangerine` K=1 subtask · 100 demos

Instruction: Pick up the tangerine.

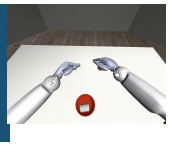
Scoring: Pick up the tangerine **(100)**.



8. `put_hotdog_bun_on_tray` K=2 subtasks · 100 demos

Instruction: Pick up the hotdog bun and place it on the tray.

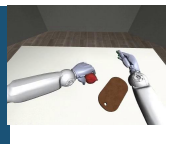
Scoring: Pick up the hotdog_bun **(50)** · Place the hotdog_bun on the tray **(50)**†.



9. `put_marshmallow_and_cookie_dough_ball_in_bowl` K=3 subtasks · 100 demos

Instruction: Put the marshmallow and the cookie dough ball in the bowl.

Scoring: Pick up the marshmallow **(25)** · Put the marshmallow in the bowl **(25)**† · Put the cookie dough ball in the bowl **(50)**†.



10. `put_onion_on_cutting_board` K=2 subtasks · 100 demos

Instruction: Pick up the onion and place it on the cutting board.

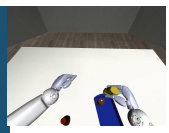
Scoring: Pick up the onion **(50)** · Place the onion on the cutting_board **(50)**†.



11. `put_peach_in_fruit_bowl` K=2 subtasks · 100 demos

Instruction: Pick up the peach and put it in the fruit bowl.

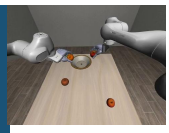
Scoring: Pick up the peach **(50)** · Place the peach in the fruit_bowl **(50)**†.



12. `place_condiments_on_tray` K=4 subtasks · 100 demos

Instruction: Place the salt and pepper shaker, mustard, and bbq sauce on the tray.

Scoring: Pick up the salt and pepper shaker **(20)** · Place the salt and pepper shaker on the tray **(20)**† · Place the mustard on the tray **(30)**† · Place the bbq sauce on the tray **(30)**†.



13. `put_apple_and_orange_in_bowl` K=6 subtasks · 100 demos

Instruction: Put one apple and one orange into the bowl, and leave one apple and one orange untouched on the table.

Scoring: Pick up the first apple **(20)** · Place the first apple into the bowl **(30)**† · Pick up the second orange **(20)** · Place the second orange into the bowl **(30)**† · Leave the first orange untouched on the table **(0)**‡ · Leave the second apple untouched on the table **(0)**‡.



14. `cook_egg_in_frypan` K=5 subtasks · 100 demos

Instruction: Pick up the frypan, then put the egg onto the frypan, and finally place the fork on the plate.

Scoring: Pick up the frypan **(20)** · Pick up the egg **(15)** · Put the egg on the frypan **(15)**† · Pick up the fork **(30)** · Place the fork on the plate **(20)**†.



15. `pick_up_spoon_from_tray_into_cup` K=2 subtasks · 100 demos

Instruction: From the tray of mixed utensils, pick up the one target spoon and drop it into the cup.

Scoring: Pick up the target utensil and drop it into the cup **(100)**† · Another utensil is not picked up **(0)**‡.



16. `grill_steak_and_corn` K=6 subtasks · 100 demos

Instruction: Open the grill lid, put the steak and the corn on the grill, then close the grill lid.

Scoring: Open the grill lid **(20)** → Pick up the steak **(20)** → Put the steak on the grill **(15)**† → Pick up the corn **(15)** → Put the corn on the grill **(15)**† → Close the grill lid **(15)**†.



17. `hang_mugs_on_tree` K=4 subtasks · 100 demos

Instruction: Pick up each of the two mugs and hang them both on the wooden mug tree.

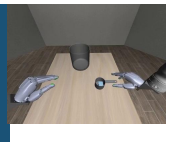
Scoring: Pick up the first mug **(10)** · Hang the first mug on the mug tree **(40)**† · Pick up the second mug **(10)** · Hang the second mug on the mug tree **(40)**†.



18. `heat_sandwich_bread_in_microwave` K=4 subtasks · 100 demos

Instruction: Open the microwave, put the sandwich bread inside, and close the microwave.

Scoring: Open the microwave door **(30)** → Pick up the sandwich bread **(20)** → Put the sandwich bread inside the microwave **(20)** → Close the microwave door **(30)**.



19. `put_measuring_cup_and_ice_into_glass` K=3 subtasks · 100 demos

Instruction: Place the ice cube in the measuring cup, then place the measuring cup and ice into the glass.

Scoring: Pick up the measuring cup **(25)** → Pick up an ice cube **(25)** → Place the measuring cup and ice into the glass **(50)**†.



20. `put_peeler_and_pizza_cutter_in_drainer` K=4 subtasks · 100 demos

Instruction: Pick up the peeler and the pizza cutter and put them in the drainer.

Scoring: Pick up the pizza cutter **(20)** · Put the pizza cutter in the drainer **(30)**† · Pick up the peeler **(20)** · Put the peeler in the drainer **(30)**†.



21. `insert_spatula_and_rolling_pin_into_glass_cup` K=6 subtasks · 100 demos

Instruction: Pick up the glass cup, then insert the spatula into the cup, insert the rolling pin into the cup, and place the cup on the placemat.

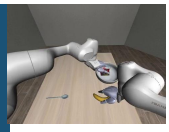
Scoring: Pick up the glass cup **(15)** · Place the glass cup on the placemat **(15)**† · Pick up the spatula **(15)** · Insert the spatula into the glass cup **(20)**† · Pick up the rolling pin **(15)** · Insert the rolling pin into the glass cup **(20)**†.



22. `load_three_canned_food_onto_dish_rack` K=6 subtasks · 100 demos

Instruction: Pick up the three cans of food and set each onto a tier of the wooden dish rack to dry.

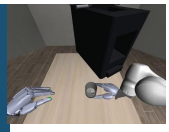
Scoring: Pick up the first can **(10)** · Set the first can onto the dish rack **(20)**† · Pick up the second can **(10)** · Set the second can onto the dish rack **(25)**† · Pick up the third can **(10)** · Set the third can onto the dish rack **(25)**†.



23. `make_cereal_in_bowl` K=6 subtasks · 100 demos

Instruction: Pick up the cereal box and put it into the red bowl, then add the banana, and place the spoon inside the bowl.

Scoring: Pick up the cereal box **(15)** · Put the cereal into the red bowl **(15)**† · Pick up the banana **(15)** · Put the banana into the red bowl **(20)**† · Pick up the spoon **(15)** · Put the spoon inside the red bowl **(20)**†.



24. `make_espresso` K=4 subtasks · 100 demos

Instruction: Put the sugar cube into the cup, then place the cup in the espresso machine's front slot.

Scoring: Pick up the sugar cube **(25)** · Drop the sugar cube into the cup **(25)**† · Pick up the cup **(25)** · Place the cup in the espresso machine front slot **(25)**†.



25. `measure_two_paprika_shakers_on_digital_scale` K=6 subtasks · 100 demos

Instruction: Take the paprika shakers one at a time, put each on the digital scale, and then remove it.

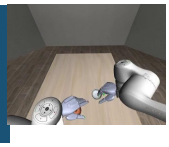
Scoring: Pick up the first paprika shaker **(10)** → Place the first paprika shaker on the digital scale **(20)** → Pick up the second paprika shaker **(20)** → Place the second paprika shaker on the digital scale **(10)** → Take the first paprika shaker off the digital scale **(20)**† → Take the second paprika shaker off the digital scale **(20)**†.



26. `organize_items_in_desk_caddy` K=6 subtasks · 100 demos

Instruction: Pick up the scissors and put them into the desk caddy, then place the soap into the desk caddy, and put the mug tree into the caddy.

Scoring: Pick up the scissors **(15)** · Put the scissors into the desk caddy **(20)**† · Pick up the soap **(15)** · Put the soap into the desk caddy **(15)**† · Pick up the mug tree **(15)** · Place the mug tree into the desk caddy **(20)**†.



27. `put_apple_in_white_bowl` K=2 subtasks · 100 demos

Instruction: Pick up the apple and put it inside the white bowl.

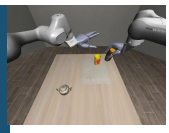
Scoring: Pick up the apple **(50)** · Put the apple inside the white bowl **(50)**†.



28. `prepare_salad_in_bowl` K=6 subtasks · 100 demos

Instruction: Place the bowl on the cutting board, then put the tomato and the lemon inside the bowl.

Scoring: Pick up the black bowl **(15)** · Place the black bowl on the cutting board **(15)**† · Pick up the tomato **(15)** · Put the tomato inside the bowl **(20)**† · Pick up the lemon **(15)** · Put the lemon inside the bowl **(20)**†.



29. `prepare_tea_with_cup` K=6 subtasks · 100 demos

Instruction: Pick up the cup and place it on the placemat, then put the honey bottle next to the cup, and place the teapot on the placemat.

Scoring: Pick up the cup **(15)** · Place the cup on the placemat **(15)**† · Pick up the honey bottle **(15)** · Place the honey bottle on the placemat **(20)**† · Pick up the teapot **(15)** · Place the teapot on the placemat **(20)**†.



30. `rethread_bagel_and_donut_onto_middle_rod` K=4 subtasks · 100 demos

Instruction: Move the bagel from the left rod to the middle rod, then the donut from the right rod to the middle rod.

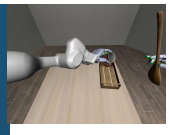
Scoring: Left arm lifts the bagel off the left rod **(25)** → Left arm re-threads the bagel onto the middle rod **(25)**† → Right arm lifts the donut off the right rod **(25)** → Right arm re-threads the donut onto the middle rod **(25)**†.



31. `place_cake_tongs_and_mug_on_tiered_shelf` K=6 subtasks · 100 demos

Instruction: Place the cake, the tongs, and the mug on the tiered shelf.

Scoring: Pick up the cake **(15)** · Place the cake on the tiered shelf **(15)**† · Pick up the mug **(15)** · Place the mug on the tiered shelf **(20)**† · Pick up the tongs **(15)** · Place the tongs on the tiered shelf **(20)**†.



32. `place_coffee_mug_and_spoon_on_tray` K=4 subtasks · 100 demos

Instruction: Place the red coffee mug and the wooden spoon on the wooden tray.

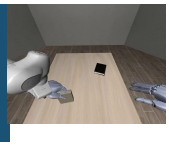
Scoring: Pick up the red coffee mug **(25)** · Place the red coffee mug on the wooden tray **(25)**† · Pick up the wooden spoon **(25)** · Put the wooden spoon on the wooden tray **(25)**†.



33. `stack_three_canned_food` K=4 subtasks · 100 demos

Instruction: Stack the three cans on top of each other.

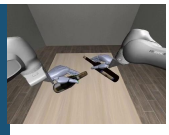
Scoring: Pick up the middle can **(10)** · Stack the middle can on the base can **(40)**† · Pick up the top can **(10)** · Stack the top can on the middle can **(40)**†.



34. `stack_yellow_book_on_black_book` K=2 subtasks · 100 demos

Instruction: Pick up the yellow book and stack it on top of the black book.

Scoring: Pick up the yellow book **(40)** · Stack the yellow book on the black book **(60)**†.



35. `stand_two_wine_bottles_upright` K=4 subtasks · 100 demos

Instruction: Pick up the two wine bottles lying on their sides and stand them upright on the table.

Scoring: Pick up the first wine bottle **(25)** → Stand the first wine bottle upright **(25)**† → Pick up the second wine bottle **(25)** → Stand the second wine bottle upright **(25)**†.



36. `store_condiments_in_short_cabinet` K=6 subtasks · 100 demos

Instruction: Open the short cabinet drawer, put the ketchup inside, put the BBQ sauce inside, then close the drawer.

Scoring: Open the short cabinet drawer **(15)** → Pick up the ketchup **(15)** → Put the ketchup inside the short cabinet **(20)** → Pick up the BBQ sauce **(15)** → Put the BBQ sauce inside the short cabinet **(20)** → Close the short cabinet drawer **(15)**.



37. `store_snacks_in_slide_cabinet` K=6 subtasks · 100 demos

Instruction: Open the slide cabinet, put the popcorn inside, put the chips inside, then close the slide cabinet.

Scoring: Open the slide cabinet door **(15)** → Pick up the popcorn **(15)** → Put the popcorn inside the slide cabinet **(20)**† → Pick up the chips **(15)** → Put the chips inside the slide cabinet **(20)**† → Close the slide cabinet door **(15)**.



38. `take_milk_from_microwave` K=3 subtasks · 100 demos

Instruction: Open the microwave, take out the milk, and place it on the table.

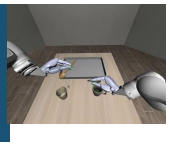
Scoring: Open the microwave door **(40)** · Pick up the milk **(30)** · Place the milk on the table **(30)**.



39. `put_three_apples_into_fruit_bowl` K=1 subtask · 100 demos

Instruction: Pick up the three apples and place them into the fruit bowl.

Scoring: Place the item into the container **(100)**†.



40. `drop_two_spoons_from_tray_into_two_cups` K=2 subtasks · 100 demos

Instruction: With each arm pick up a spoon from the tray and drop it, one each, into the two cups.

Scoring: Drop the first spoon into a cup **(50)**† $\rightarrow$ Drop the second spoon into a cup **(50)**†.

B. Instruction-Following (20 specifications, 20 with released demonstrations)

Verbal-command fidelity, tested three ways: *count* variants share one scene but demand different object counts, *combo* tasks conjoin two instructions, and *switch* tasks replace the instruction mid-episode through a language schedule, so the policy must serve the current command rather than a remembered one. For switch tasks the two instruction phases are listed as Phase 1 and Phase 2; grasping a wrong object terminates the episode.



1. `hammer_one_nail_each_hand` K=4 subtasks · 100 demos

Instruction: With the left arm use the left hammer to drive the up nail into the board, and with the right arm use the right hammer to drive the down nail.

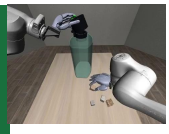
Scoring: Left arm drives the up nail into the board **(25)** · Right arm drives the down nail into the board **(25)** · Extra left nail is not driven into the board **(25)**† · Extra right nail is not driven into the board **(25)**†.



2. `hammer_two_nails_each_hand` K=4 subtasks · 100 demos

Instruction: With the left arm use the left hammer to drive the two left nails into the board, and with the right arm use the right hammer to drive the two right nails.

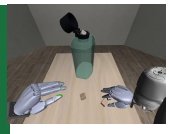
Scoring: Left arm drives the first left nail into the board **(25)** · Left arm drives the second left nail into the board **(25)** · Right arm drives the first right nail into the board **(25)** · Right arm drives the second right nail into the board **(25)**.



3. `open_bottle_and_drop_one_sugar_cube_in_bottle` K=9 subtasks · 100 demos

Instruction: With the left arm flip open the bottle's articulated cap, and with the right arm pick up the first sugar cube and drop it into the bottle.

Scoring: Left arm flips open the bottle's articulated cap **(10)** · Right arm picks up the first sugar cube **(10)** · Right arm drops the first sugar cube into the bottle **(20)** · Extra sugar cube 1 never enters the bottle **(10)**‡ · Extra sugar cube 2 never enters the bottle **(10)**‡ · Extra sugar cube 3 never enters the bottle **(10)**‡ · Extra sugar cube 1 is never picked up **(10)**‡ · Extra sugar cube 2 is never picked up **(10)**‡ · Extra sugar cube 3 is never picked up **(10)**‡.



4. `open_bottle_and_drop_two_sugar_cubes_in_bottle` K=9 subtasks · 100 demos

Instruction: With the left arm flip open the bottle's articulated cap, and with the right arm pick up the first and second sugar cubes and drop them into the bottle.

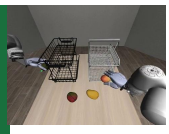
Scoring: Left arm flips open the bottle's articulated cap **(10)** · Right arm picks up the first sugar cube **(10)** · Right arm drops the first sugar cube into the bottle **(15)** · Right arm picks up the second sugar cube **(10)** · Right arm drops the second sugar cube into the bottle **(15)** · Extra sugar cube 1 never enters the bottle **(10)**‡ · Extra sugar cube 2 never enters the bottle **(10)**‡ · Extra sugar cube 1 is never picked up **(10)**‡ · Extra sugar cube 2 is never picked up **(10)**‡.



5. `pack_four_mangoes_into_baskets` K=8 subtasks · 100 demos

Instruction: With the left arm pick up the two leftmost mangoes and place them into the left basket, and with the right arm pick up the two rightmost mangoes and place them into the right basket.

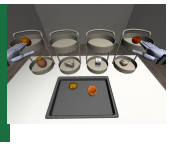
Scoring: Left arm picks up the first (leftmost) mango **(10)** · Left arm places the first mango into the left basket **(20)**† · Left arm picks up the second mango **(10)** · Left arm places the second mango into the left basket **(10)**† · Right arm picks up the fourth (rightmost) mango **(10)** · Right arm places the fourth mango into the right basket **(20)**† · Right arm picks up the third mango **(10)** · Right arm places the third mango into the right basket **(10)**†.



6. `pack_two_mangoes_into_two_baskets` K=10 subtasks · 100 demos

Instruction: With the left arm pick up the leftmost mango and place it into the left basket, and with the right arm pick up the rightmost mango and place it into the right basket.

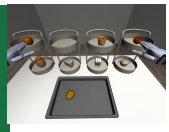
Scoring: Left arm picks up the first (leftmost) mango **(10)** · Left arm places the leftmost mango into the left basket **(10)**† · Right arm picks up the fourth (rightmost) mango **(10)** · Right arm places the rightmost mango into the right basket **(10)**† · Extra mango_1 stays out of the left basket **(10)**‡ · Extra mango_1 stays out of the right basket **(10)**‡ · Extra mango_2 stays out of the left basket **(10)**‡ · Extra mango_2 stays out of the right basket **(10)**‡ · Extra mango_1 is not picked up **(10)**‡ · Extra mango_2 is not picked up **(10)**‡.



7. `put_jam_and_lemon_from_tray_onto_shelf` K=2 subtasks · 30 demos

Instruction: Take the jam and the lemon out of the tray and place them back onto the shelf.

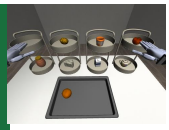
Scoring: Put the jam back onto the shelf **(50)**† · Put the lemon back onto the shelf **(50)**†.



8. `put_lemon_on_shelf_then_take_jam_into_tray` K=2 subtasks · 30 demos

Instruction (switches after control step 782)*: Phase 1:* Take the lemon out of the tray and place it onto the shelf. *Phase 2:* Leave the lemon. Pick up the jam from the shelf and put it into the tray instead.

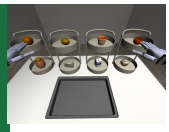
Scoring: Put the lemon onto the shelf **(50)** → Put the jam into the tray **(50)**†.



9. `put_orange_on_shelf_then_take_back_into_tray` K=2 subtasks · 30 demos

Instruction (switches after control step 750)*: Phase 1:* Take the orange out of the tray and place it onto the shelf. *Phase 2:* Put the orange back into the tray.

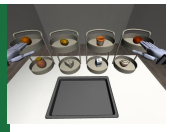
Scoring: Put the orange onto the shelf **(50)** → Put the orange back into the tray **(50)**†.



10. `take_jam_into_tray_then_put_back_on_shelf` K=2 subtasks · 30 demos

Instruction (switches after control step 674)*: Phase 1:* Pick up the jam from the shelf and put it into the tray. *Phase 2:* Put the jam back onto the shelf.

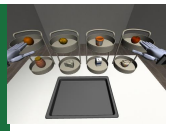
Scoring: Put the jam into the tray **(50)** → Put the jam back onto the shelf **(50)**†.



11. `take_milk_and_lemon_from_shelf_into_tray` K=2 subtasks · 30 demos

Instruction (switches after control step 492)*: Phase 1:* Pick up the milk carton from the shelf and put it into the tray. *Phase 2:* Leave the milk. Pick up the lemon from the shelf and put it into the tray instead.

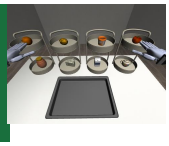
Scoring: Put the milk carton into the tray **(50)** → Put the lemon into the tray **(50)**†.



12. `take_teapot_into_tray_then_put_back_on_shelf` K=2 subtasks · 30 demos

Instruction (switches after control step 536)*: Phase 1:* Pick up the teapot from the shelf and put it into the tray. *Phase 2:* Put the teapot back onto the shelf.

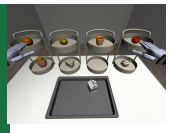
Scoring: Put the teapot into the tray **(50)** → Put the teapot back onto the shelf **(50)**†.



13. `take_jam_and_lemon_from_shelf_into_tray` K=2 subtasks · 30 demos

Instruction: Pick up the jam and the lemon from the shelf and put them into the tray.

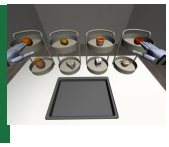
Scoring: Put the jam into the tray **(50)**† · Put the lemon into the tray **(50)**†.



14. `take_jam_into_tray_and_put_milk_on_shelf` K=2 subtasks · 30 demos

Instruction: Pick up the jam from the shelf and put it into the tray, and take the milk carton out of the tray and place it back onto the shelf.

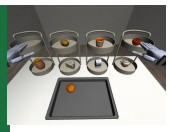
Scoring: Put the jam into the tray **(50)**† · Put the milk carton back onto the shelf **(50)**†.



15. `take_milk_and_orange_from_shelf_into_tray` K=2 subtasks · 30 demos

Instruction: Pick up the milk carton and the orange from the shelf and put them into the tray.

Scoring: Put the milk carton into the tray **(50)**† · Put the orange into the tray **(50)**†.



16. `take_teapot_into_tray_and_put_orange_on_shelf` K=2 subtasks · 30 demos

Instruction: Pick up the teapot from the shelf and put it into the tray, and take the orange out of the tray and place it onto the shelf.

Scoring: Put the teapot into the tray **(50)**† · Put the orange onto the shelf **(50)**†.



17. `thread_bagel_and_donut_onto_middle_and_side_rods` K=4 subtasks · 100 demos

Instruction: Thread the bagel onto the middle rod and the donut onto the left rod.

Scoring: Left arm picks up the donut **(10)** · Left arm threads the donut onto the left rod **(40)**† · Right arm picks up the bagel **(10)** · Right arm threads the bagel onto the middle rod **(40)**†.



18. `thread_bagel_and_donut_onto_side_rods` K=4 subtasks · 100 demos

Instruction: Thread the bagel onto the right rod and the donut onto the left rod.

Scoring: Right arm picks up the donut **(10)** · Right arm threads the donut onto the left rod **(40)**† · Left arm picks up the bagel **(10)** · Left arm threads the bagel onto the right rod **(40)**†.



19. `water_one_plant_each_hand` K=4 subtasks · 100 demos

Instruction: Use the left watering can to water the leftmost plant and the right watering can to water the rightmost plant.

Scoring: Right arm waters the rightmost plant **(25)** · Left arm waters the leftmost plant **(25)** · Extra plant_1 is not watered **(25)**‡ · Extra plant_2 is not watered **(25)**‡.



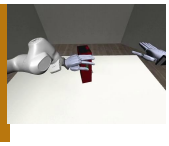
20. `water_two_plants_each_hand` K=4 subtasks · 100 demos

Instruction: Use the left watering can to water the two left plants and the right watering can to water the two right plants.

Scoring: Right arm waters the first right plant **(25)** · Right arm waters the second right plant **(25)** · Left arm waters the first left plant **(25)** · Left arm waters the second left plant **(25)**.

C. Limited-Demonstrations (26 specifications, 26 with released demonstrations)

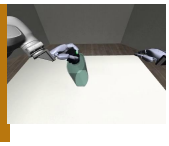
Data-scarce skill acquisition: every task ships with exactly 30 demonstrations. The axis includes ten articulated-mechanism tasks, such as opening a microwave, cash register, or rice cooker, which require force-aware actuation of hinged, sliding, and rotary joints. This regime makes 44-DoF hand coordination decisive.



1. `pull_down_slot_machine_lever` K=1 subtask · 30 demos

Instruction: Pull down the lever of the slot machine.

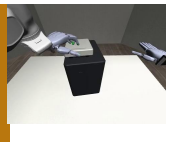
Scoring: Pull down the lever of the slot machine **(100)**.



2. `flip_open_bottle_cap` K=1 subtask · 30 demos

Instruction: Flip open the bottle's articulated cap.

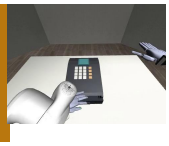
Scoring: Flip open the bottle's articulated cap **(100)**.



3. `open_bread_maker_lid` K=1 subtask · 30 demos

Instruction: Open the lid of the bread maker.

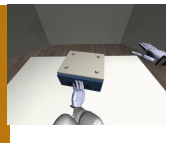
Scoring: Open the lid of the bread maker **(100)**.



4. `pull_open_cash_register_drawer` K=1 subtask · 30 demos

Instruction: Pull open the cash register drawer.

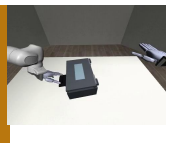
Scoring: Pull open the cash register drawer **(100)**.



5. `pull_open_cabinet_drawer` K=1 subtask · 30 demos

Instruction: Pull open the drawer of the cabinet.

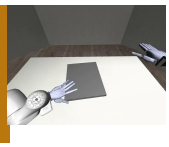
Scoring: Pull open the drawer of the cabinet **(100)**.



6. `open_flatbed_scanner_lid` K=1 subtask · 30 demos

Instruction: Open the lid of the flatbed scanner.

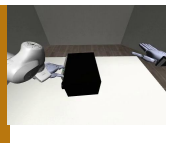
Scoring: Open the lid of the flatbed scanner **(100)**.



7. `raise_laptop_stand_platform` K=1 subtask · 30 demos

Instruction: Raise the platform of the laptop stand.

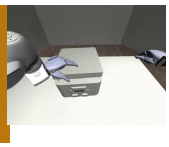
Scoring: Raise the platform of the laptop stand **(100)**.



8. `open_microwave_oven_door` K=1 subtask · 30 demos

Instruction: Open the microwave oven door.

Scoring: Open the microwave oven door **(100)**.



9. `open_rice_cooker_lid` K=1 subtask · 30 demos

Instruction: Open the lid of the rice cooker.

Scoring: Open the lid of the rice cooker **(100)**.



10. `open_wheelie_bin_lid` K=1 subtask · 30 demos

Instruction: Open the lid of the wheelie bin.

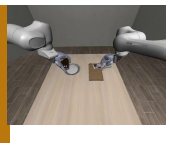
Scoring: Open the lid of the wheelie bin **(100)**.



11. `assemble_arch_bridge` K=6 subtasks · 30 demos

Instruction: Stand two piers on the base, lay the deck across them, then set the capstone cube on the middle of the deck.

Scoring: Stand pier A on the base **(10)**† · Stand pier B on the base **(10)**† · Pick up the deck **(20)** · Lay the deck across the two piers **(20)**† · Pick up the capstone **(20)** · Set the capstone on the deck centre **(20)**†.



12. `arrange_bar_on_tray_and_cake_on_plate` K=4 subtasks · 30 demos

Instruction: Place the food bar on the tray, then place the cake on the plate.

Scoring: Pick up the food bar **(20)** · Place the food bar on the tray **(30)**† · Pick up the cake **(20)** · Place the cake on the plate **(30)**†.



13. `arrange_three_bagels_in_oven_tray` K=6 subtasks · 30 demos

Instruction: Pick up the three bagels one at a time and lay each flat in the oven tray so it rests within the tray.

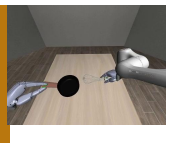
Scoring: Pick up the first bagel **(10)** · Lay the first bagel in the oven tray **(20)**† · Pick up the second bagel **(10)** · Lay the second bagel in the oven tray **(25)**† · Pick up the third bagel **(10)** · Lay the third bagel in the oven tray **(25)**†.



14. `put_apple_and_orange_into_basket` K=4 subtasks · 30 demos

Instruction: Pick up the apple and the orange and put them both inside the basket.

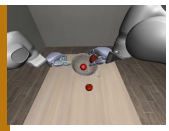
Scoring: Pick up the apple **(25)** · Put the apple inside the basket **(25)**† · Pick up the orange **(25)** · Put the orange inside the basket **(25)**†.



15. `pick_up_whisk` K=1 subtask · 30 demos

Instruction: Pick up the whisk.

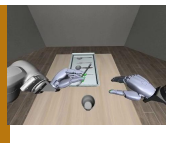
Scoring: Pick up the whisk **(100)**†.



16. `drain_three_tomatoes_in_strainer` K=7 subtasks · 30 demos

Instruction: Pick up the three tomatoes and place them into the strainer so they rest in it.

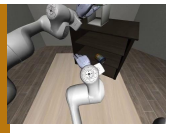
Scoring: Pick up the strainer **(10)**† · Pick up the first tomato **(10)** · Place the first tomato into the strainer **(20)**† · Pick up the second tomato **(10)** · Place the second tomato into the strainer **(20)**† · Pick up the third tomato **(10)** · Place the third tomato into the strainer **(20)**†.



17. `pick_up_fork_from_tray_into_cup` K=2 subtasks · 30 demos

Instruction: From the tray of mixed utensils, pick up the one fork and drop it into the cup.

Scoring: Pick up the target utensil and drop it into the cup **(100)**† · Another utensil is not picked up **(0)**‡.



18. `insert_items_into_shelf` K=6 subtasks · 30 demos

Instruction: Pick up the alphabet soup and the BBQ sauce and insert them into the wooden shelf, then place the white storage box on the shelf.

Scoring: Pick up the alphabet soup **(15)** · Insert the alphabet soup into the shelf **(20)**† · Pick up the BBQ sauce **(15)** · Insert the BBQ sauce into the shelf **(20)**† · Pick up the storage box **(15)** · Place the storage box on the shelf **(15)**†.



19. `rotate_two_cubes_around_plates` K=5 subtasks · 30 demos

Instruction: Move the two cubes clockwise around the three triangle plates one move at a time, always moving the cube before the empty plate onto it, until each returns to its start.

Scoring: Right arm moves the cube before the empty plate onto it **(50)** → Left arm moves the cube before the empty plate onto it **(30)** → Left arm moves the last cube back to its start **(20)**† → Second cube restored on its own plate **(0)**† → The plate that started empty is empty again **(0)**†.



20. `pack_three_avocados_into_tiered_basket` K=6 subtasks · 30 demos

Instruction: Pick up the three avocados one at a time and drop them into the tiered basket so they rest in it.

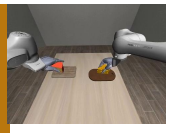
Scoring: Pick up the first avocado **(10)** · Place the first avocado into the basket **(20)**† · Pick up the second avocado **(10)** · Place the second avocado into the basket **(25)**† · Pick up the third avocado **(10)** · Place the third avocado into the basket **(25)**†.



21. `place_two_oranges_on_shelf_tiers` K=2 subtasks · 30 demos

Instruction: Pick up the two oranges and place one on the lower tier of the shelf and the other on the upper tier.

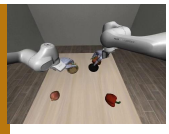
Scoring: Place one orange on the upper tier of the shelf **(40)**† · Place the other orange on the lower tier of the shelf **(60)**†.



22. `serve_cheese_on_two_cutting_boards` K=4 subtasks · 30 demos

Instruction: Pick up a cheese block and set it flat on its cutting board, one cheese per board.

Scoring: Pick up the first cheese block **(25)** · Set the first cheese block on its cutting board **(25)**† · Pick up the second cheese block **(25)** · Set the second cheese block on its cutting board **(25)**†.



23. `sort_pepper_and_onion_into_two_bowls` K=4 subtasks · 30 demos

Instruction: Put one bell pepper into one bowl and one onion into the other, keeping each kind separate.

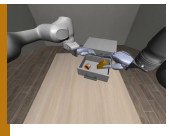
Scoring: Pick up the first bell pepper **(25)** · Place the first bell pepper into its bowl **(25)**† · Pick up the first onion **(25)** · Place the first onion into its bowl **(25)**†.



24. `store_frozen_food_in_freezer` K=6 subtasks · 30 demos

Instruction: Open the chest freezer lid, put the ice cubes and boxed food inside, then close the freezer lid.

Scoring: Open the chest freezer lid **(20)** → Pick up the ice cubes **(20)** → Put the ice cubes inside the freezer **(20)**† → Pick up the boxed food **(15)** → Put the boxed food inside the freezer **(20)**† → Close the freezer lid **(5)**.



25. `store_items_in_white_cabinet` K=8 subtasks · 30 demos

Instruction: Open the white cabinet middle drawer, put the cheese and the jam inside, then close the middle drawer and open the top drawer.

Scoring: Open the white cabinet middle drawer **(20)** → Pick up the cheese **(10)** → Put the cheese inside the white cabinet **(15)**† → Pick up the jam **(10)** → Put the jam inside the white cabinet **(15)**† → Close the white cabinet middle drawer **(15)** → Open the white cabinet top drawer **(15)** → Close the white cabinet top drawer **(0)**.



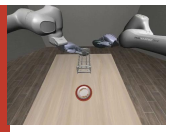
26. `put_three_eggs_into_tupperware` K=1 subtask · 30 demos

Instruction: Pick up the three eggs one at a time and place them into the tupperware.

Scoring: Place the item into the container **(100)**†.

D. Long-Horizon (10 specifications, 10 with released demonstrations)

Multi-stage chores, including barbecue grilling, fridge restocking, dessert plating, and recycling sorting, with 4 to 9 scored subtasks and 6.5 objects per scene on average. Their sampled demonstrations average 48.3 s. Success requires sustaining a plan far beyond the policy's action chunk.



1. `insert_three_plates_into_dish_rack` K=6 subtasks · 100 demos

Instruction: Pick up the three plates and insert each into the dish rack to dry.

Scoring: Pick up the first plate **(10)** · Insert the first plate into the dish rack **(20)**† · Pick up the second plate **(10)** · Insert the second plate into the dish rack **(25)**† · Pick up the third plate **(10)** · Insert the third plate into the dish rack **(25)**†.



2. `grill_drumsticks_fish_and_squash` K=7 subtasks · 100 demos

Instruction: Open the grill lid, place the two drumsticks, the two fish, and the squash on the grill, then close the lid.

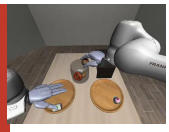
Scoring: Left arm opens the grill lid **(15)** → Left arm lays the first drumstick on the grill **(15)**† → Left arm lays the second drumstick on the grill **(15)**† → Right arm lays the first fish on the grill **(15)**† → Right arm lays the second fish on the grill **(15)**† → Right arm lays the squash on the grill **(15)**† → Left arm closes the grill lid **(10)**†.



3. `fill_pot_and_bottle_with_sugar_cubes_and_veg` K=9 subtasks · 100 demos

Instruction: Left arm opens the bottle's cap, drops the two vegetables and the largest sugar cube into the pot, closes the cap, and lifts it, while the right arm drops the other three sugar cubes into the bottle.

Scoring: Left arm opens the bottle's articulated cap **(20)** → Left arm places the first vegetable into the pot **(10)**† → Left arm places the second vegetable into the pot **(10)**† → Left arm drops the largest sugar cube into the pot **(10)**† → Right arm drops the second sugar cube into the bottle **(10)**† → Right arm drops the third sugar cube into the bottle **(10)**† → Right arm drops the fourth sugar cube into the bottle **(10)**† → Left arm lifts the pot **(10)**† → Left arm closes the bottle's articulated cap **(10)**†.



4. `fill_jar_with_snacks` K=8 subtasks · 100 demos

Instruction: Take the jar off the shelf and set it on the table, put all the snacks into the jar, and place the jar back on the shelf.

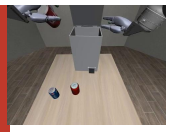
Scoring: Take the jar off the shelf **(20)** → Place the first donut into the jar **(10)**† → Place the second donut into the jar **(10)**† → Drops the first marshmallow into the jar **(10)**† → Place the first cupcake into the jar **(10)**† → Place the second cupcake into the jar **(10)**† → Drop the second marshmallow into the jar **(10)**† → Place the jar back on the shelf **(20)**†.



5. `restock_fridge_with_condiments_and_bottles` K=7 subtasks · 100 demos

Instruction: Open the fridge, put the two condiments and the two tall bottles inside, and close the door with the water bottle standing upright.

Scoring: Left arm opens the fridge door **(25)** → Right arm puts the water bottle into the fridge **(10)**† → Right arm puts the bottled drink into the fridge **(10)**† → Left arm puts the mayonnaise into the fridge **(10)**† → Left arm puts the mustard into the fridge **(10)**† → Left arm closes the fridge door **(15)**† → The water bottle stands upright in the fridge **(20)**†.



6. `recycle_five_cans_into_bin` K=7 subtasks · 100 demos

Instruction: Open the pedal bin lid, toss the five cans in, then close the lid.

Scoring: Left arm flips open the bin lid **(15)** → Left arm tosses the first can into the bin **(15)**† → Left arm tosses the second can into the bin **(15)**† → Left arm tosses the third can into the bin **(15)**† → Right arm tosses the fourth can into the bin **(15)**† → Right arm tosses the fifth can into the bin **(15)**† → Left arm closes the bin lid **(10)**†.



7. `sort_mixed_food_into_boxes` K=6 subtasks · 100 demos

Instruction: Put the mushrooms into the middle vegetable basket, the hot dogs into the front meat basket, and the drink cartons into the back drinks basket.

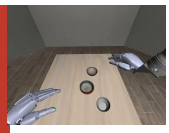
Scoring: Drop the first mushroom into the middle vegetable basket **(10)**† · Drop the second mushroom into the middle vegetable basket **(10)**† · Drop the first hot dog into the front meat basket **(20)**† · Drop the second hot dog into the front meat basket **(20)**† · Drop the first drink carton into the back drinks basket **(20)**† · Drop the second drink carton into the back drinks basket **(20)**†.



8. `place_strainer_with_tomatoes_and_spoon_on_dish_rack` K=8 subtasks · 100 demos

Instruction: Pick up the strainer, put the two tomatoes into it, lift it onto the dish rack, then place the spoon onto the dish rack as well.

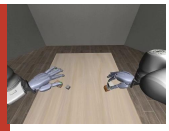
Scoring: Pick up the strainer **(10)** · Pick up the first tomato **(10)** · Place the first tomato into the strainer **(10)**† · Pick up the second tomato **(10)** · Place the second tomato into the strainer **(10)**† · Place the strainer onto the dish rack **(15)**† · Pick up the spoon **(15)** · Place the spoon onto the dish rack **(20)**†.



9. `nest_four_bowls` K=3 subtasks · 100 demos

Instruction: Nest these four bowls of decreasing sizes into a stack, fitting each smaller bowl inside the larger one beneath it.

Scoring: Nest the 12 cm bowl into the 13 cm base bowl **(20)**† · Nest the 11.5 cm bowl into the 12 cm bowl **(30)**† · Nest the 10 cm bowl into the 11.5 cm bowl **(50)**†.



10. `stack_sugar_cubes` K=4 subtasks · 100 demos

Instruction: With the left arm stack three sugar cubes on top of each other, and with the right arm stack three sugar cubes on top of each other.

Scoring: Left arm stacks the middle cube on the bottom cube **(25)**† · Left arm stacks the top cube on the middle cube **(25)**† · Right arm stacks the middle cube on the bottom cube **(25)**† · Right arm stacks the top cube on the middle cube **(25)**†.

E. Memory (10 specifications, 10 with released demonstrations)

Latent-state tracking under perceptual aliasing: in *color-swap cups* a cube's color is visually removed after a button sequence and must be recalled to choose the matching cup, in *picnic-basket FIFO* removal order must match insertion order, and in *memory cups* a reveal phase shows target contents that are later hidden. These tasks are impossible from a single frame and probe genuine temporal state tracking rather than reactive control.



1. `drop_fruits_in_basket_then_remove_in_order` K=2 subtasks · 100 demos

Instruction: With the left arm drop the three fruits into the basket in any order, then with the right arm take them back out in the same order they went in.

Scoring: Left arm places all three objects into the basket **(50)** → Right arm removes all three in the order they were placed **(50)**†.



2. `swap_cup_colors_with_cube` K=4 subtasks · 100 demos

Instruction: With the right arm press the yellow button three times, pick up the cube and swing it over the two cups five times to swap their colors, then place it in the matching-color cup.

Scoring: Right arm presses the yellow button exactly three times, the cube's color disappears **(25)**† → Pick up the cube **(5)** → Pass over the cups so the two cups swap colors five times **(25)** → Place the cube into the cup whose color matches it **(45)**†.



3. `drop_items_into_picnic_basket_then_remove_in_order` K=2 subtasks · 100 demos

Instruction: Drop the hotdog bun, baguette, juice carton and water bottle into the picnic basket in any order, then take them out in the same order.

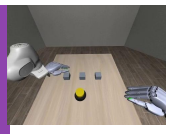
Scoring: Both arms place all four items into the basket **(50)** → Both arms remove all four items in the same order they were placed **(50)**†.



4. `uncover_cubes_in_order` K=7 subtasks · 100 demos

Instruction: Press the yellow button three times to cover the three cubes with caps, then lift the caps to uncover the red, green, and blue cubes in that order.

Scoring: Right arm presses the yellow button, the caps appear **(10)**† → All three cubes are covered by the caps **(0)** → Lift the cap covering the red cube first **(30)** → Then lift the cap covering the green cube **(30)** → Finally lift the cap covering the blue cube, all revealed **(30)**† → Red cube still uncovered at the end **(0)**† → Green cube still uncovered at the end **(0)**†.



5. `uncover_red_cube` K=5 subtasks · 100 demos

Instruction: Press the yellow button three times to lift only the cover concealing the red cube.

Scoring: Right arm presses the yellow button exactly three times, the caps appear **(20)**† → All three cubes are covered by the caps **(0)** → Lift only the cap covering the red cube **(80)**† → Green cube still covered at the end **(0)**† → Blue cube still covered at the end **(0)**†.



6. `pour_ice_between_two_cups` K=2 subtasks · 100 demos

Instruction: Right arm pours the ice from the right cup into the left cup, then left arm pours it back into the right cup and sets both cups down.

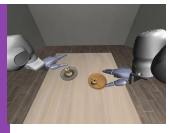
Scoring: Right arm pours the ice from the right cup into the left cup **(50)** → Left arm pours the ice back into the right cup and sets both cups down **(50)**†.



7. `press_button_then_grasp_mug` K=3 subtasks · 100 demos

Instruction: With the right arm press the yellow button exactly four times, and only then, with the left arm pick up the mug and place it on the tray.

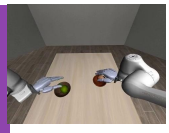
Scoring: Right arm presses the yellow button exactly four times **(60)**† → Left arm picks up the mug **(10)** → Left arm places the mug on the tray **(30)**†.



8. `swap_candles_between_bowls` K=5 subtasks · 100 demos

Instruction: Swap and restore the two candles.

Scoring: Left arm moves the left candle to the right bowl **(20)** → Right arm moves the right candle to the left bowl **(20)** → Left arm moves the right candle back to the right bowl **(30)** → Right arm moves the left candle back to the left bowl (restored) **(30)**† → Right candle restored to the right bowl **(0)**†.



9. `swap_fruits_between_bowls` K=4 subtasks · 100 demos

Instruction: Left arm moves the peach to the right bowl, right arm moves the lime to the left, then left arm moves the lime to the right and right arm moves the peach to the left.

Scoring: Left arm moves the peach to the right bowl **(25)** → Right arm moves the lime to the left bowl **(25)** → Left arm moves the lime back to the right bowl **(25)** → Right arm moves the peach back to the left bowl **(25)**†.



10. `press_red_button_then_blue_button` K=2 subtasks · 100 demos

Instruction: With the left arm press the red button exactly ten times, and only then, with the right arm press the blue button once.

Scoring: Left arm presses the red button exactly ten times **(50)**† · Right arm presses the blue button once **(50)**†.

VIII. Discussion and Limitations

The five-axis organization reflects a deliberate benchmark design. As VLA training data grows, average-case skill becomes less discriminative, while memory, horizon, instruction fidelity, data efficiency, and robustness remain capability edges. Progress-based predicate scoring makes the heterogeneous tasks commensurable, exposes partial progress when strict success is not reached, and keeps evaluation fully automated. The resulting per-axis vectors show whether short-horizon competence extends to memory and long-horizon tasks. ID/OOD gaps measure aggregate robustness under the combined scripted shifts, and Limited-Demonstrations isolates acquisition from 30 examples.

BiDex-100 is a rigid-body benchmark. MuJoCo does not model fluids, deformable objects, or rich frictional contact at the fidelity of hardware, and sim-to-real transfer of policies trained on the released demonstrations remains to be validated on the physical platforms. The object library, while diverse, is centered on tabletop food and kitchen scenes. The OOD shifts are scripted and finite; learned or adversarially generated shift distributions are a natural extension.

Within this scope, evaluating seven contemporary VLA models under one protocol establishes reference baselines and a diagnostic framework of capability profiles, ID/OOD robustness gaps, and low-data regimes. We release the code, tasks, data, and evaluation tooling to support reproducible community use.